Enhancing Continual Semantic Segmentation via Uncertainty and Class Balance Re-Weighting

Zichen Liang, Yusong Hu, Fei Yang, and Xialei Liu

***Abstract*—Continual Semantic Segmentation (CSS) primarily aims to continually learn new semantic segmentation categories while avoiding catastrophic forgetting. In semantic segmentation tasks, images can comprise both familiar old categories and novel unseen categories and they are treated as background in the incremental stage. Therefore, it is necessary to utilize the old model to generate pseudo-labels. However, the quality of these pseudo-labels significantly influences the model's forgetting of the old categories. Erroneous pseudo-labels can introduce harmful gradients, thus exacerbating model forgetting. In addition, the issue of class imbalance poses a significant challenge within the realm of CSS. Although traditional methods frequently diminish the emphasis placed on new classes to address this imbalance, we discover that the imbalance extends beyond the distinction between old and new classes. In this paper, we specifically address two previously overlooked problems in CSS: the impact of erroneous pseudo-labels on model forgetting and the confusion induced by class imbalance. We propose an Uncertainty and Class Balance Re-weighting approach (UCB) that assigns higher weights to pixels with pseudo-labels exhibiting lower uncertainty and to categories with smaller proportions during the training process. Our proposed approach enhances the impact of essential pixels during the continual learning process, thereby reducing model forgetting and dynamically balancing category weights based on the dataset. Our method is simple yet effective and can be applied to any method that uses pseudo-labels. Extensive experiments on the Pascal-VOC and ADE20K datasets demonstrate the efficacy of our approach in improving model performance across three state-of-the-art methods. The code will be available at https://github.com/JACK-Chen-2019/UCB**

***Index Terms*—Continual learning, semantic segmentation, uncertainty.**

I. Introduction

THE advent of deep learning has significantly enhanced the performance of semantic segmentation [1], [2], [3], [4], [5], [6]. This fundamental task is widely employed across various industries and plays a crucial role in complex applications. However, the training of semantic segmentation networks requires a large amount of data and computational resources. Furthermore, when new data or classes are introduced, the model often has to restart the learning process from scratch. As a solution, the concept of Continual Semantic Segmentation (CSS) has been proposed, which requires the model to continually learn new classes while retaining knowledge of previously learned ones.

In traditional CSS settings, the model can only obtain labels for new classes, while any other objects are considered as background. Consequently, the model is susceptible to overfitting on the new classes and forgetting the old knowledge, which is also known as catastrophic forgetting [7]. Most of the existing methods primarily focus on alleviating catastrophic forgetting by designing model structures [8], [9] and employing distillation techniques [10], [11]. Due to the lack of labels for the previous classes, existing methods utilize the old model to generate pseudo-labels [11], [12], [13], [14] for the old categories in the training set, aiming to alleviate forgetting and reduce semantic shift between background and old classes.

However, as continual learning progresses, the model gradually forgets the old classes, making these pseudo-labels increasingly unreliable. Moreover, using these erroneous labels further exacerbates forgetting in the model, leading to a sharp decline in model performance. Historically, previous approaches have shown limited attention to the quality of pseudo-labels. Most methods have not taken any countermeasures or simply filtered out unreliable samples based on a threshold [11], [14]. However, relying solely on filtering is inadequate as samples with higher reliability may still be incorrect. We argue that the quality of pseudo-labels plays a crucial role in continual learning. As illustrated in Fig. 1, we use entropy as a measure of uncertainty. It is observed that even when entropy is low, the model still has a probability of making incorrect predictions, despite the relatively low error rate. These erroneous pseudo-labels can also have adverse effects on the model. Considering that uncertainty can reflect the quality of pseudo-labels, we can assign varied weights to pixels based on their levels of uncertainty, where higher weights are assigned to samples with lower uncertainty. This approach allows high-confidence labels to have a greater impact during gradient backpropagation, thereby reducing the negative effects of low-confidence labels that are likely to be incorrect.

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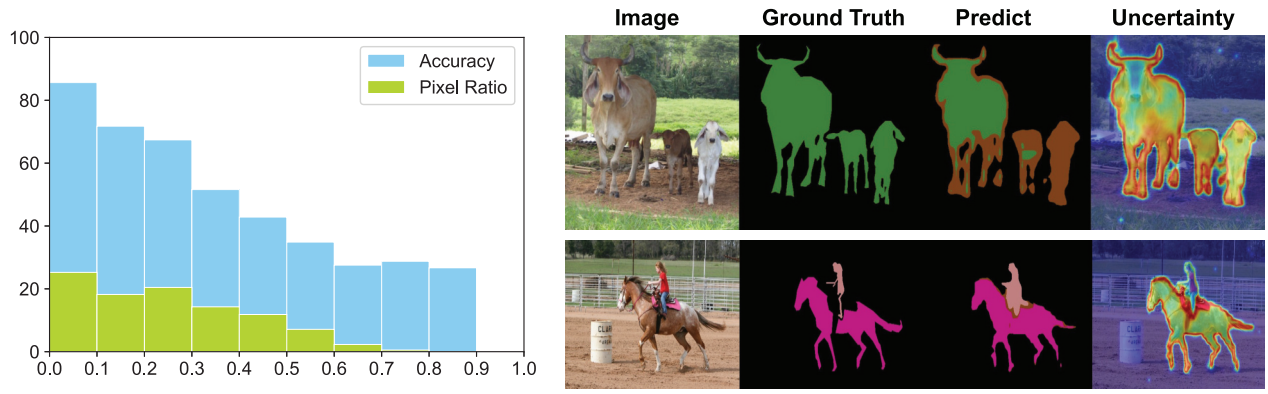
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(a) Uncertainty and accuracy

(b) Uncertainty visualization

Fig. 1. **The relationship between uncertainty and accuracy.** Figure (a) presents the statistical analysis of the relationship between prediction accuracy and uncertainty of PLOP in the 15-1 task. The uncertainty is categorized into 10 bins, and the accuracy (in blue) and pixel ratio (in green) of foreground categories are computed for each bin. Higher uncertainty leads to lower accuracy. Figure (b) visually illustrates the uncertainty, with the red regions representing higher uncertainty and a higher likelihood of prediction errors.

In addition, class imbalance poses a significant challenge in CSS. Consider the scenario in our daily life where we aim to train a model on new categories. In this scenario, we annotate only the new classes in an image and provide it to the model for learning. CSS also follows a similar scenario, where images containing new categories are selected from an incremental dataset and only the new categories are annotated. Consequently, this unavoidably results in the new classes dominating the dataset in most conditions, particularly in the case of single-class per-step tasks. Previous work [12], [15] mitigates this problem by reducing the weight of new categories. However, due to the randomness of real-world scenarios, class imbalance occurs not only between new and old categories but also among the old categories themselves and among the new categories. Therefore, simply adjusting the balance between new and old categories cannot straightforwardly solve the problem.

To address the aforementioned two issues, we propose **U**ncertainty and **C**lass **B**alance Re-weighting approach (**UCB**) based on pixel-wise loss re-weighting. This strategy dynamically allocates the weight based on the quality of generated pseudo-labels and the balance of categories in the training set. In particular, we utilize entropy to measure the reliability of pseudo-labels and adjust the weights of different reliability levels using an exponential function. Additionally, we employ the calculation of pixel proportion per class as an indicator of class equilibrium. The class balance weight is determined by contrasting this proportion against the ideal class ratio. Therefore, classes with higher proportions will be assigned lower weights.

In summary, the main contributions of this paper are as follows:

- We perform an in-depth theoretical analysis to assess the quality of pseudo-labels and tackle the issue of class imbalance within CSS sceneries.
- We propose UCB, which integrates uncertainty-aware weights with exponential decay and class-specific balance weights. UCB effectively mitigates catastrophic forgetting caused by erroneous pseudo-labels and better adapts to the more complex class balance scenarios in CSS.
- The proposed method is general and effective, demonstrating performance improvement across diverse methods and settings.

II. Related Work

A. Continual Learning

In recent years, deep learning has made significant advances in the three major visual tasks: classification [16], object detection [17], and segmentation [18]. However, most of them require retraining when incorporating new data or new classes. Therefore, continual learning primarily addresses how models can continuously learn new knowledge. Its main challenge is catastrophic forgetting [7], where the model experiences significant forgetting of previously learned knowledge after acquiring new knowledge without retraining.

Currently, continual learning has gained extensive research attention and made significant strides in the field of image classification tasks. Existing methods can be classified into regularization-based methods, structure-based methods, and replay-based methods. Regularization-based methods, using distillation [19], [20], [21], [22], [23] or through imposing constraints on important parameters in the model [19], [24], [25], [26], [27], [28] to reduce forgetting. Structure-based methods [29], [30], [31], [32], [33] employ dynamic network architectures to alleviate forgetting. Replay-based methods allow the model to store a small portion of old data [22], [34], [35], [36], [37], [38] or utilize an auxiliary model [39], [40], [41] to generate synthetic data and utilize them with new-class data for updating the current model. Additionally, BMC [42] utilizes cross-attention mechanisms to encode the matching relationships between images and each category. GMM [43] directly generates labels for images through an adapted generative model. In dealing with real-world data that exhibit non-uniform distribution, [44] proposes a method to address class imbalance by re-weighting the gradients.

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B. Continual Semantic Segmentation(CSS)

Continual Semantic Segmentation is an extension of CIL, where the model needs to conduct pixel classification and continually learn new classes step by step.

The main challenge of CSS is catastrophic forgetting and distillation is widely used. ILT [45] introduces continual setting to semantic segmentation task and employs distillation on both output logits and intermediate features to alleviate catastrophic forgetting. PLOP [11] utilizes multi-scale pooling distillation to preserve spatial relationships across different ranges. UCD [10] considers the predict uncertainty with distillation. LSKD [46] proposes a layer-specific knowledge distillation method to allocate suitable knowledge schemes and weights to different intermediate layers.

Some methods focus on solving the drift between actual background and possible future classes. SSUL [13] utilizes additional saliency map data to effectively distinguish the actual background and latent future classes. Another work [47] employs unsupervised contrastive learning to optimize future classes by utilizing visual similarity and feature affinity. MBS [48] analyzes the causes of background shift and proposes several optimization strategies to mitigate it. BARM [49] presents a Background Adaptation strategy, optimizing an additional channel for residual modeling to prevent unstable optimization and chaotic behavior in background logits.

Replay-based methods store a small amount of previous data in memory to retain information about old classes. However, this approach incurs memory costs and raises privacy concerns. RECALL [50] utilizes generative adversarial networks and web crawling to obtain category data, thus avoiding privacy concerns. To reduce memory requirements, ALIFE [51] saves features instead of images and uses rotation matrices to update features for each category. Prior research [52] considers multiple factors such as sample diversity and category performance to perform effective replay.

Other methods consider reducing forgetting from a structural perspective. RCIL [9] utilizes representation compensation to decouple the representation learning of old knowledge and new knowledge. EWF [8] and subsequent work [53] employ dynamic endpoint weight fusion to alleviate model forgetting. More recently, NeST [54] introduces a classifier pre-tuning method that learns a transformation from old classifiers to generate new ones. CoMasTRe [55] simplifies continual segmentation by decoupling the task into objectness learning and class recognition.

C. Uncertainty Guided Pseudo-Labeling

Pseudo-labeling is widely used in Semi-supervised Learning, Domain Adaptation, and Continual Semantic Segmentation. In Semi-supervised Learning, the model needs to make the most use of unlabeled data to train the model. Domain Adaptation necessitates the model to adapt to the unlabeled target data and make necessary adjustments. In Continual Semantic Segmentation, the model requires assigning pseudo-labels to old categories against forgetting. As the incremental process proceeds, the model will gradually forget the old categories, and pseudo-labels will result in significant cumulative errors. However, currently, most methods do not have specific measures to mitigate this issue.

The reduction of the negative impact of erroneous pseudo-labels on model performance has become a key area of research, with uncertainty emerging as a crucial metric for assessing pseudo-label quality. We categorize existing uncertainty-based approaches into three primary strategies: filtering, class-specific thresholding, and the use of multiple experts.

The simplest approach is to filter out samples with high uncertainty. Early methods assess uncertainty based on the logits after applying the softmax function [56], while later methods [11] utilize entropy theory to measure uncertainty. In the domain of semantic segmentation, MBS [48] addresses the problem by considering background shift, filtering out potentially erroneous pseudo-labels by comparing the confidence levels of the foreground and background. URN [57] uses a scaling operation combined with Conditional Random Fields (CRF) to simulate the true behavior of noisy pseudo-masks. While these methods are relatively simple, their effectiveness is limited in practice due to the small proportion of high-uncertainty samples.

Class-specific thresholding takes into account the varying difficulty of learning different classes by setting distinct thresholds for each class. PLOP [11] adjusts thresholds based on the median entropy across the current dataset, while ST [58] uses two Gaussian distributions to distinguish between reliable and unreliable samples for each class. Previous work [59] dynamically adjusts thresholds based on the logits' performance for each class and re-weights the classes to address class imbalance. These methods enable threshold adjustments based on the difficulty of each class, but they assume a clear boundary between the entropy of correct and incorrect pseudo-labels. In practice, pseudo-labels with low uncertainty may still be incorrect, albeit with a low probability, making it difficult to establish a clear boundary. Additionally, these methods incur extra computational costs and training time.

Multi-expert approaches involve combining the outputs of multiple experts to assess the uncertainty of pseudo-labels. Typically, 2 to 3 or more experts are used, each providing a different perspective on outputs, making this ensemble strategy widely applicable. For instance, [60] employs an uncertainty-weighted integration strategy to obtain segmentation predictions from teacher models, while [61] leverages multiple experts to evaluate prediction uncertainty. While these methods improve the model's ability to assess prediction uncertainty, they also require training multiple expert models, which adds to the computational burden.

III. Preliminaries

A. Problem Definition

We follow the traditional definition of CSS [11], [13], [47], [62], where the model needs to continually learn new classes step by step. In step t, denoted as $t \in \{1, \ldots, T\}$, the model $f_{t,\boldsymbol{\theta}}$ is trained using parameters $\boldsymbol{\theta}$ on a sub-dataset $\mathcal{D}^t$ that includes samples from at least one of the new categories $\mathcal{C}^t$. Within the dataset $\mathcal{D}^t$, each pair $(\boldsymbol{x}^t, \boldsymbol{y}^t)$ consists of an input

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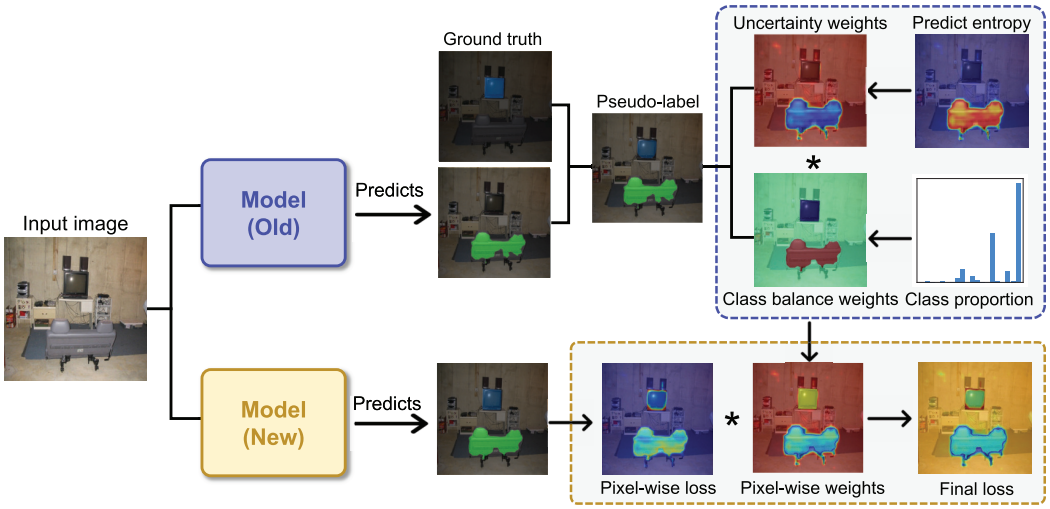


Fig. 2. **Explanation of UCB.** Our approach aims to address the impact of erroneous labels and class imbalance by considering both prediction uncertainty and class balance factors to assign different weights to individual pixels. Specifically, we utilize entropy as a measure of uncertainty, allocating higher weights to pixels with lower entropy. We accurately assess the class proportion by calculating pixel ratios for different categories and assigning class balance weights based on their proportions. Finally, we multiply the calculated pixel-wise loss by the aforementioned weights to obtain the final loss.

image $\boldsymbol{x}^t \in \mathbb{R}^{H\times W\times C}$ and its corresponding ground-truth mask $\boldsymbol{y}^t \in \mathbb{R}^{H\times W}$. In traditional CSS scenarios, the previous and unseen classes are treated as background, and their labels are uniformly represented as c_b. Therefore, in step t, the training label space is denoted as $\mathcal{C}^t \cup \{c_b\}$.

During incremental steps, the model often experiences severe catastrophic forgetting of old classes, primarily caused by the inability to acquire ground-truth labels for old classes. Existing approaches typically employ the old model $f_{t-1,\boldsymbol{\theta}}$ to generate pseudo-labels $\tilde{\boldsymbol{y}}_t$ for old classes. However, as continual learning progresses, the quality of these pseudo-labels deteriorates further. Another problem exists in the class imbalance of training data especially for single-class steps, which can result in significant class confusion.

B. Revisiting Pseudo-Labeling in CSS

As mentioned above, the incremental dataset $\mathcal{D}^t$ only contains the new categories $\mathcal{C}^t$, and old categories are labeled as background c_b. Directly using these labels can lead to background shift [62] and catastrophic forgetting [7], thus the pseudo-labeling strategy is widely used in CSS.

At step t ($t > 1$), for a given training image $\boldsymbol{x}^t$ and its label $\boldsymbol{y}^t$, we utilize the previous model f_{t-1} to obtain the old output logits $\hat{\boldsymbol{o}}^{t-1} \in \mathbb{R}^{H\times W\times|\mathcal{C}^{1:t-1}\cup c_b|}$. Then we apply the `argmax` function to retrieve the predicted labels $\hat{\boldsymbol{y}}^{t-1} \in \mathbb{R}^{H\times W}$. Next, we can obtain the final label $\tilde{\boldsymbol{y}}^t$ for classification by replacing the background regions of $\boldsymbol{y}^t$ with the corresponding regions of $\hat{\boldsymbol{y}}^{t-1}$:

$$\tilde{\boldsymbol{y}}^t = \begin{cases} \boldsymbol{y}^t & \text{where } \boldsymbol{y}^t \in \mathcal{C}^t \\ \hat{\boldsymbol{y}}^{t-1} & \text{where } \boldsymbol{y}^t = c_b. \end{cases} \tag{1}$$

The generated pseudo-labels can guide the process of supervised classification and feature distillation for old classes. In this work, we are not concerned about the specific methods used to obtain the pseudo-labels, but we emphasize the importance of the quality of the generated pseudo-labels.

IV. Method

The pipeline of UCB is illustrated in Fig. 2. We re-weight the pixel-wise loss $\mathcal{L}_{ce} \in \mathbb{R}^{H\times W}$ considering both prediction uncertainty and class imbalance:

$$\boldsymbol{w}(\boldsymbol{x}_{ij}^t) = \boldsymbol{w}^{ua}(\hat{\boldsymbol{o}}_{ij}^{t-1}) \cdot \boldsymbol{w}^{cb}(\tilde{\boldsymbol{y}}_{ij}^t), \tag{2}$$

where $\boldsymbol{x}_{ij}^t$ represents the pixel at (i, j) of the image. $\boldsymbol{w}^{ua}$ and $\boldsymbol{w}^{cb}$ refer to uncertainty-aware weights and class balance weights, respectively. These will be discussed in detail in Section IV-A and Section IV-B.

To maintain the relative proportion of classification loss alongside other losses, we normalize the weights by dividing them by their mean value, ensuring that the average weight is 1. The final classification loss can be calculated as:

$$\mathcal{L}_{clf}(\boldsymbol{x}^t) = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W}\frac{\boldsymbol{w}(\boldsymbol{x}_{ij}^t)}{\bar{\boldsymbol{w}}(\boldsymbol{x}^t)} \cdot \mathcal{L}_{ce}(\boldsymbol{o}_{ij}^t, \tilde{\boldsymbol{y}}_{ij}^t), \tag{3}$$

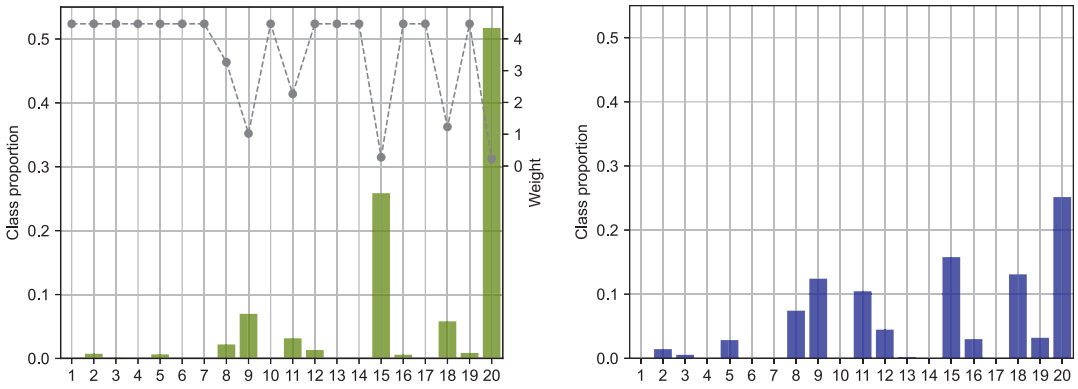
where $\boldsymbol{o}_{ij}$ represents the output of the pixel at (i, j) and $\bar{\boldsymbol{w}}(\boldsymbol{x}^t)$ represents the average class weights of $\boldsymbol{x}^t$. $\mathcal{L}_{ce}$ denotes the pixel-wise loss and can be substituted with binary cross-entropy (BCE) or other supervised loss functions.

A. Uncertainty-Aware Pseudo-Label Re-Weighting

To mitigate the adverse effects caused by these mislabeled pseudo-labels, previous methods [11], [47] use dynamic or specific thresholds to filter unconfident pixels. However, as shown in Fig. 1, the pseudo-labels which have lower entropy can also be associated with incorrect predictions, but with a lower probability. In this paper, we propose an uncertainty-aware pseudo-label re-weighting strategy to mitigate the impact of unreliable gradients caused by incorrect pseudo-labels.

At incremental step t, we can obtain the previous output logits $\hat{\boldsymbol{o}}^{t-1} \in \mathbb{R}^{H\times W\times|\mathcal{C}^{1:t-1}\cup\{c_b\}|}$ using the old model f_{t-1} and generate the pseudo-labels $\tilde{\boldsymbol{y}}^t$ using Eq. (1). The main idea is to assign different weights to pixels based on their level

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(a) Class proportion and class balance weights (b) Class proportion after balance

Fig. 3. **Class proportion visualisation.** We conduct a statistical analysis of the category pixel ratios (a) and the adjusted category pixel ratios (b) on the Pascal-VOC 15-1 overlapped task. Generally, the percentage of the new category is higher (as seen in the 20th category monitor), but there are also cases of complex imbalances (excessive percentage in the 15th category person). The category weights are depicted by the gray dashed line, with values greater than 1 indicating an increase in weight. To prevent excessive weight, we set upper and lower limits for the weight values.

of uncertainty. As the uncertainty increases, the reliability of the pseudo-labels decreases, thus necessitating a smaller weight allocation to mitigate the potential impact of erroneous gradients.

Following traditional methods, we use entropy to identify the pseudo-labels' uncertainty. It can be calculated as:

$$E(\hat{\boldsymbol{o}}_{ij}^{t-1}) = \frac{1}{N_{old}} \sum_{c=1}^{N_{old}} \hat{\boldsymbol{o}}_{ijc}^{t-1} \cdot \log(\hat{\boldsymbol{o}}_{ijc}^{t-1}), \tag{4}$$

where $N_{old} = |\mathcal{C}^{1:t-1} \cup \{c_b\}|$ represents the total number of previous classes. We set the entropy of new class pixels to zero since they are labeled, thus having zero uncertainty.

Then we can assign varying weights to pixels based on their respective degrees of uncertainty. Motivated by the work of [63], we employ exponential decay to establish the weighting scheme.

$$\boldsymbol{w}^{ua}(\hat{\boldsymbol{o}}_{ij}^{t-1}) = e^{-\frac{\lambda}{E_0} \cdot \left[E(\hat{\boldsymbol{o}}_{ij}^{t-1}) - E_0\right]} \cdot \mathbb{I}_{\{E(\hat{\boldsymbol{o}}_{ij}^{t-1}) < E_0\}}, \tag{5}$$

where $\mathbb{I}_{\{\cdot\}}$ represents an indicator, and E_0 is a predefined threshold used for filtering. To mitigate the impact of E_0 on the weight magnitudes, we employ the scaling factor $\frac{\lambda}{E_0}$ to dynamically allocate weights within the range $[1, \lambda e]$, assigning a weight of 1 when entropy equals E_0 and a weight of λe when entropy equals 0. The E_0 and λ are hyperparameters and will be discussed in Section V-C.

B. Pixel-Wise Class Balance Re-Weighting

As mentioned above, CSS faces a more prominent issue of class imbalance, especially when learning tasks with multiple steps and single-class steps, where the number of pixels belonging to new classes is significantly higher than that of old classes. The imbalance between new and old categories can lead to the model overfitting to new classes and misclassifying similar old classes as new ones. As a result, the performance of the model is significantly impaired. Except for the imbalance between new and old classes, as shown in Fig. 3, the imbalance between different old classes can also be large. Moreover, in practical applications, the state of class balance can become highly intricate. To fundamentally address this issue, it is imperative to consider the balance of all classes. Therefore, we propose a pixel-wise class balance re-weighting strategy to solve this problem.

For each step, we employ an additional vector $\boldsymbol{b}^t \in \mathbb{R}^{|\mathcal{C}^{1:t}|}$ to capture the pixel proportion for each class, which can directly and effectively reflect the balance between classes. Specifically, we compute the frequency of pixel $\boldsymbol{h}(\tilde{\boldsymbol{y}}^t) \in \mathbb{R}^{|\mathcal{C}^{1:t}|}$ of the generated pseudo-labels $\tilde{\boldsymbol{y}}^t$ within a batch, and then calculate the class proportion $\boldsymbol{p}^t$.

As training progresses, we continuously update the overall vector. We keep a record of the total number of encountered samples, denoted as N, and use it as a coefficient for the sliding average update. This process is formulated as:

$$\boldsymbol{b}^t = \begin{cases} \boldsymbol{p}^t & \text{if } N = 0 \\ \frac{N}{N+B}\boldsymbol{b}^t + \frac{B}{N+B}\boldsymbol{p}^t & \text{if } N > 0, \end{cases} \tag{6}$$

where B represents the current batch size, and N is updated to $N + B$ after finish this batch.

We use $\frac{1}{|\mathcal{C}^{1:t}|}$ as the ideal balanced class proportion and compare it with the class proportion $\boldsymbol{b}^t$ to determine the weight coefficients for each class. By dividing the recorded class proportion $\boldsymbol{b}^t$ by the ideal class proportion $\frac{1}{|\mathcal{C}^{1:t}|}$, weights that are higher than the ideal value will be reduced to less than 1, whereas weights lower than the ideal value will be increased beyond 1. The class balance weight α^t for each class can be calculated as:

$$\alpha_c^t = \text{clamp}(\frac{1}{|\mathcal{C}^{1:t}| \cdot \boldsymbol{b}_c^t}, \min, \max), \tag{7}$$

where the `clamp` function is used to limit the weights within the range $[\min, \max]$. By default, following [12], min is set to $\sqrt{\frac{C^t}{C^{1:t-1}}}$ and max is set to $\sqrt{\frac{C^{1:t-1}}{C^t}}$.

Next, we can obtain the class balance weights $\boldsymbol{w}^{cb}(\tilde{\boldsymbol{y}}_{ij}^t)$ by allocating the coefficients obtained from Eq. (7) based on the class of each pixel. The overall algorithm of UCB in incremental steps can be found in Algorithm 1.

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Algorithm 1 Pseudo Code for UCB in Incremental Steps

Require: old model f_{t-1}, dataset $(\boldsymbol{x}^t, \boldsymbol{y}^t)$, total steps T

$t \leftarrow 1$
while $t \leq T$ **do**
  Initialize the class proportion vector $\boldsymbol{b}^t$
  while not converged **do**
    Sample data pairs $(\boldsymbol{x}^t, \boldsymbol{y}^t)$
    Get pseudo-labels $\tilde{\boldsymbol{y}}^t$ using Eq. (1)
    Calculate uncertainty E using Eq. (4)
    Get uncertainty-aware weights $\boldsymbol{w}^{ua}$ using Eq. (5)
    Update the class proportion vector $\boldsymbol{b}^t$ using Eq. (6)
    Get class balance weights w^{cb} using Eq. (7)
    Get re-weighted loss $\mathcal{L}_{clf}$ using Eq. (3)
  end while
  $t \leftarrow t + 1$
end while

C. Theoretical Motivation and Analysis

To simplify the theoretical analysis, we can transform the segmentation problem into a simplified binary segmentation task, where the model needs to classify the image pixel $\boldsymbol{x}$ into two categories: $+1$ represents the positive class, labeled as $\boldsymbol{y}^+$, and -1 represents the negative class, labeled as $\boldsymbol{y}^-$. We assume that the features of the positive and negative samples follow two independent Gaussian distributions:

$$\boldsymbol{x}|\boldsymbol{y}^+ \sim \mathcal{N}(\boldsymbol{\mu}_+, \sigma^2), \boldsymbol{x}|\boldsymbol{y}^- \sim \mathcal{N}(\boldsymbol{\mu}_-, \sigma^2). \tag{8}$$

Ideally, the number of positive and negative samples in the dataset is balanced, i.e., $n^+ = n^-$. Under this assumption, we can easily derive the following Bayesian classifier:

$$f(\boldsymbol{x}) = \text{sign}\left(\boldsymbol{x} - \frac{\boldsymbol{\mu}_+ + \boldsymbol{\mu}_-}{2}\right). \tag{9}$$

By comparing $\boldsymbol{x}$ with $\frac{\boldsymbol{\mu}_+ + \boldsymbol{\mu}_-}{2}$, we can obtain the classification result.

In the CSS scenario, the old classes need to be labeled by the old model, which may lead to incorrect pseudo-labels. Additionally, the class imbalance is exacerbated in CSS scenarios, especially in the single-class scenario. Here, we assume that both y^+ and y^- belong to the old classes and examine how incorrect pseudo-labels influence the accuracy of the estimated values.

1) Considering Wrong Pseudo-Labels: Suppose that the probability of correctly predicting a positive sample $\boldsymbol{x}_i^+$ is p_i^+, so the probability of predicting this positive sample as a negative sample is $1 - p_i^+$. Considering the presence of incorrect pseudo-labels, the estimated mean value for positive samples is given by:

$$\tilde{\boldsymbol{\mu}}_+ = \frac{\sum_{i=1}^{n^+} p_i^+ \boldsymbol{\mu}_+ + \sum_{i=1}^{n^+}(1 - p_i^+)\boldsymbol{\mu}_-}{n^+}. \tag{10}$$

Assuming that $\mu_- > \mu_+$, we can compute the difference between the estimated $\tilde{\boldsymbol{\mu}}_+$ and the true $\boldsymbol{\mu}_+$ as follows:

$$|\tilde{\boldsymbol{\mu}}_+ - \boldsymbol{\mu}_+| = \frac{\sum_{i=1}^{n^+}(1 - p_i^+)(\boldsymbol{\mu}_- - \boldsymbol{\mu}_+)}{n^+}. \tag{11}$$

As p_i^+ approaches 1, the estimated value converges to the true value. As p_i^+ approaches 0, the estimated mean converges to $\boldsymbol{\mu}_-$, and the model's performance on positive samples decreases. To mitigate the impact of these estimate errors, we can assign a weight to each sample based on its predicted accuracy. A straightforward approach is to multiply each sample by a weight $w(p_i) = (p_i)^\lambda$, where $\lambda > 0$. Higher values of λ diminish the impact of samples with lower predicted accuracy. In practice, many methods [11], [56], [61] estimate the accuracy of pseudo-labels using the entropy of outputs logits $\boldsymbol{o}$. According to our statistics, as shown in Fig. 1(a), we assume that entropy and accuracy follow an exponential decay relationship. Consequently, we model the probability p_i as $p_i \sim e^{-E(\boldsymbol{o}_i)}$, where $E(\boldsymbol{o})$ denotes the entropy of the outputs logits. Based on this, we also design the weight to decay exponentially with entropy, which can be expressed as $w(\boldsymbol{o}) \sim e^{-\lambda E(\boldsymbol{o})}$, as shown in Eq. (5).

2) Considering Class Imbalance: We can estimate the decision boundary between the positive and negative classes using the following equation:

$$\hat{\theta} = \frac{1}{2}(\tilde{\mu}_+ + \tilde{\mu}_-) = \frac{1}{2}\left(\frac{\sum_{i=1}^{n^+} \boldsymbol{x}_i^+}{n^+} + \frac{\sum_{i=1}^{n^-} \boldsymbol{x}_i^-}{n^-}\right). \tag{12}$$

Based on the theory from [64], for any $\delta > 0$, with probability at least $1 - 2e^{-\frac{2\delta^2}{9\sigma^2}\cdot\frac{\tilde{n}_+\tilde{n}_-}{\tilde{n}_+ + \tilde{n}_-}} - 2e^{-\frac{8\tilde{n}_+\delta^2}{9(\mu_+ - \mu_-)^2}} - 2e^{-\frac{8\tilde{n}_-\delta^2}{9(\mu_+ - \mu_-)^2}}$, the error between our estimates $\hat{\theta}$ and the excepted values $\frac{\mu_+ + \mu_-}{2}$ is less than δ. This implies that data imbalance influences the probability of obtaining such a good estimation. Note that $\frac{\tilde{n}_+\tilde{n}_-}{\tilde{n}_+ + \tilde{n}_-}$ is maximized when $\tilde{n}_+ = \tilde{n}_-$. Ideally, we aim for the number of positive samples to equal the number of negative samples, i.e., $n^+ = n^-$. A straightforward approach to achieve this is to adjust the number of samples from different classes, ensuring that the distribution of sample sizes is as balanced as possible. For deep learning networks, this can be accomplished by adjusting the sample weights. Consequently, we calculate the proportion of each class and assign appropriate weights to balance the classes as much as possible, ensuring $n^+ = n^-$. Naturally, the above conclusion is not limited to the old classes and can be generalized to all classes in continual learning, as demonstrated in Eq. (7).

V. Experiments

In this section, we will introduce the experimental setup and training details. We conduct experiments by applying our method to state-of-the-art methods that utilize pseudo-labeling: PLOP [1][11], SSUL [2][13] and CoinSeg [3][14] (based on Transformer). We conducted comprehensive quantitative and qualitative experiments based on this foundation.

A. Experiments Setup

1) Datasets and Metrics: We conduct extensive experiments on Pascal-VOC [65] and ADE20K [66] datasets. The Pascal-VOC dataset consists of 10,582 training images and

[1]https://github.com/arthurdouillard/CVPR2021_PLOP
[2]https://github.com/clovaai/SSUL
[3]https://github.com/zkzhang98/CoinSeg

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TABLE I

COMPARISON OF MIOU(%) AT THE FINAL STEP ON THE PASCAL-VOC DATASET VARIES ACROSS DIFFERENT INCREMENTAL SCENARIOS. THE * DENOTES RESULTS REPORTED IN THE ORIGINAL PAPER, WHEREAS † REPRESENTS RESULTS REPRODUCED BY US. JOINT IS THE UPPER BOUND

Method	Backbone	19-1 (2 steps)			15-5 (2 steps)			15-1 (6 steps)			10-1 (11 steps)		
		0-19	20	all	0-15	16-20	all	0-15	16-20	all	0-10	11-20	all
ILT* [45]	Res101	67.75	10.88	65.05	67.08	39.23	60.45	8.75	7.99	8.56	7.20	3.70	5.50
MiB* [62]	Res101	71.43	23.59	69.15	76.37	49.97	70.08	34.22	13.50	29.29	12.25	13.09	12.65
SDR* [67]	Res101	69.10	32.60	67.40	57.40	52.60	69.90	44.70	21.80	39.20	32.40	17.10	25.10
RCIL* [9]	Res101	77.00	31.50	74.70	78.80	52.20	72.40	70.60	23.70	59.40	55.40	15.10	34.30
SPPA* [68]	Res101	76.50	36.20	74.60	78.10	52.90	72.10	66.20	23.30	56.00	-	-	-
RBC* [15]	Res101	77.26	55.60	76.23	76.59	52.78	70.92	69.54	38.44	62.14	-	-	-
ALIFE* [51]	Res101	76.61	49.36	75.31	77.18	52.52	71.31	64.44	34.91	57.41	-	-	-
MiB+EWF* [8]	Res101	77.80	12.20	74.70	-	-	-	78.00	25.50	65.50	56.00	16.70	37.30
AMSS* [52]	Res101	79.40	42.80	77.66	79.31	55.88	73.73	78.54	50.82	71.94	-	-	
CoMasTRe* [55]	Res101	75.13	69.51	74.86	79.73	51.93	73.11	69.77	43.62	63.54	-	-	-
MiB+NeST* [54]	Res101	71.70	28.20	69.70	71.70	28.20	69.70	61.70	20.40	51.80	52.30	21.00	37.40
BARM* [49]	Res101	78.20	42.20	76.40	-	-	-	77.60	45.90	70.00	72.20	51.00	62.10
PLOP† [11]	Res101	76.36	29.47	74.12	75.13	47.84	68.63	67.30	13.11	54.40	44.03	15.51	30.45
+NeST [54]	Res101	77.00	49.10	75.70	77.60	55.80	72.40	72.20	33.70	63.10	54.20	17.80	36.90
+UCB (Ours)	Res101	76.76	32.48	74.65	76.33	51.09	70.32	71.83	15.43	58.40	50.76	20.36	36.28
SSUL† [13]	Res101	77.71	29.58	75.42	77.41	50.42	70.97	76.39	39.74	67.66	72.17	45.89	59.66
+UCB (Ours)	Res101	78.05	39.32	76.20	78.59	51.73	72.20	78.57	43.14	70.13	75.18	45.46	61.03
Joint*	Res101	81.00	79.10	80.90	82.70	75.00	80.90	82.70	75.00	80.90	82.10	79.60	80.90
CoinSeg† [14]	Swin-B	81.14	37.70	79.07	83.29	67.00	79.41	81.73	55.30	75.44	77.98	61.48	70.13
+UCB (Ours)	Swin-B	81.69	60.43	80.68	83.27	70.18	80.15	83.28	57.45	77.14	79.72	65.28	72.84
Joint*	Swin-B	82.60	84.40	82.70	83.80	79.30	82.70	83.80	79.30	82.70	82.40	83.00	82.70

1,449 test images, covering 20 foreground categories. In the incremental process, we encounter significant class imbalance issues in the Pascal-VOC dataset. ADE20K dataset comprises 20,210 training images and 2,000 test images, representing 150 different categories. Due to the larger number of categories, the ADE20K dataset has a relatively balanced distribution during the incremental process. We utilize mean Intersection-over-Union (mIoU) as our evaluation metric and computed the mIoU for both base classes $\boldsymbol{C}^1 \cup \{c_b\}$ and the incremental classes $\boldsymbol{C}^{2:T}$ as indicators of forgetting and plasticity.

2) Incremental Protocols: The training process of CSS is divided into T steps, with each step representing a task. At every step, the model is tasked with learning the newly introduced categories, which are different from the categories learned in prior steps. Following common methods [8], [11], [13], we adopt the overlapped setting which is more realistic. In this setting, the background class may encompass both the old categories and unknown new categories.

Following standard protocols, we utilize various segmentation benchmarks for Pascal-VOC. These include 15-1(6 steps, where the first step trains 15 categories and each subsequent step trains one additional category), 15-5 (2 steps), 19-1 (2 steps), as well as evaluation on more challenging scenarios such as 10-1 (11 steps). For ADE20K, we evaluated our approach on 100-50 (2 steps), 100-10 (6 steps), 100-5(11 steps), and 50-50 (3 steps).

3) Implementation Details: We use identical parameters including the learning rate, batch size, optimizer, and epochs as the source code framework, with the only alteration being the weighting process for the loss function. When calculating the class balance weights, we set the class weight of the background class to 1, while the remaining classes follow the computation method outlined in Eq. (7). In determining the ideal balance weights, only the classes that have been encountered are taken into consideration. Through the observations on several experimental setups of the ADE20K dataset, we found that the class distribution is relatively more balanced compared to Pascal-VOC. Therefore, we narrow the range for class imbalance adjustments, setting the min and max values in Eq. (7) to 0.8 and 1.25, respectively. All experiments are conducted on 2 GeForce RTX 3090 GPUs. More specifically, in PLOP, we use a batch size of 12 and train for 30 epochs per task. The learning rate is set to 0.01 for the base task and 0.001 for the incremental tasks, employing the DeeplabV3 architecture with a ResNet-101 backbone. For SSUL, we train for 75 epochs with a batch size of 16 and a learning rate of 0.01, also using the DeeplabV3 architecture with a ResNet-101 backbone. For CoinSeg, the learning rate is set to 0.0001, with a batch size of 12 for Pascal VOC 2012 and 6 for ADE20K, training for 50 epochs. The DeeplabV3 network is used for segmentation, with the Swin-B model as the backbone.

B. Evaluation Results

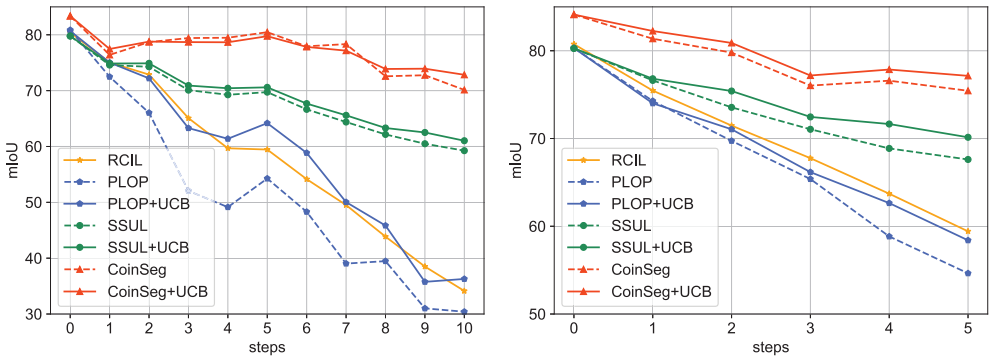
1) Comparison on Pascal-VOC: We apply our method to PLOP [11], SSUL [2][13] and CoinSeg [14] on the Pascal-VOC dataset. As illustrated in Table I and Table III, our

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TABLE II

COMPARISON OF MIOU(%) AT THE FINAL STEP ON THE ADE20K DATASET VARIES ACROSS DIFFERENT INCREMENTAL SCENARIOS. THE * DENOTES RESULTS REPORTED IN THE ORIGINAL PAPER, WHEREAS † REPRESENTS RESULTS REPRODUCED BY US. JOINT IS THE UPPER BOUND

Method	Backbone	100-50 (2 steps)			100-10 (6 steps)			100-5 (11 steps)			50-50 (3 steps)		
		old	new	all	old	new	all	old	new	all	old	new	all
MiB* [62]	Res101	40.52	17.17	32.79	38.21	11.12	29.24	36.01	5.66	25.96	45.57	21.01	29.31
SDR [67]	Res101	37.40	24.80	33.20	28.90	7.40	21.70	-	-	-	40.90	23.80	29.50
ALIFE* [51]	Res101	42.18	23.07	35.86	41.02	22.76	34.98	-	-	-	48.98	25.69	33.56
RCIL* [9]	Res101	42.30	18.80	34.50	39.50	17.70	32.10	38.50	11.50	29.60	48.30	25.00	32.50
SPPA* [68]	Res101	42.90	19.90	35.20	41.00	12.50	31.50	-	-	-	49.80	23.90	32.50
AMSS* [52]	Res101	44.06	24.96	37.74	43.88	25.14	37.67	43.35	18.53	35.13	-	-	-
MiB+EWF* [8]	Res101	41.20	21.30	34.60	41.50	-	33.20	41.40	13.40	32.10	-	-	-
CoMasTRe* [55]	Res101	45.73	26.02	39.20	42.32	18.42	34.41	40.82	15.83	32.55	-	-	-
MiB+NeST* [54]	Res101	40.30	24.60	35.10	40.20	20.60	33.70	39.90	18.00	32.70	45.60	26.80	33.20
BARM* [49]	Res101	42.00	23.00	35.70	41.10	23.10	35.20	40.50	21.20	34.10	47.90	26.50	33.70
PLOP† [11]	Res101	41.66	15.42	32.97	38.95	14.98	30.93	35.88	8.24	26.73	47.31	20.83	29.80
+NeST [54]	Res101	42.20	24.30	36.30	40.90	22.00	34.70	39.30	17.40	32.00	48.70	27.70	34.80
+UCB (Ours)	Res101	41.95	21.13	35.05	38.47	19.47	32.18	35.54	13.14	28.13	47.32	24.60	32.27
SSUL† [13]	Res101	40.25	19.85	33.49	39.50	19.90	33.01	39.17	18.30	32.27	46.20	19.96	28.82
+UCB (Ours)	Res101	40.70	21.74	34.42	39.96	20.80	33.62	39.77	19.22	32.96	46.67	23.21	31.13
Joint*	Res101	43.80	28.90	38.90	43.80	28.90	38.90	43.80	28.90	38.90	50.70	32.80	38.90
CoinSeg† [14]	Swin-B	41.60	26.70	36.60	42.29	24.77	36.49	42.48	24.20	36.43	48.49	29.01	35.59
+UCB (Ours)	Swin-B	43.09	26.43	37.58	42.94	25.14	37.05	43.07	23.59	36.63	49.32	29.89	36.45
Joint*	Swin-B	43.50	30.60	39.20	43.50	30.60	39.20	43.50	30.60	39.20	50.20	33.70	39.20



(a) 10-1 scenario on Pascal-VOC (b) 15-1 scenario on Pascal-VOC

Fig. 4. Comparison of mIoU(%) at each step for 10-1 (a) and 15-1 (b) settings. The dashed line represents the baseline, while the solid line of the corresponding color represents our method. Initially, our method exhibits marginal differences from the baseline in the early stages but progressively outperforms the baseline, indicating its effectiveness in mitigating the forgetting issue.

method has demonstrated notable success in these three methods, particularly excelling on PLOP with more challenging task 10-1 (11 steps). Our approach not only alleviates the forgetting of old categories but also effectively enhances the learning of new categories in many scenarios, indicating its capability to strengthen model adaptability. While NeST [54] performs well on PLOP, it necessitates additional computational cost—specifically, five extra epochs for pre-tuning the new classifier for each incremental task, whereas our method requires minimal extra computation and demonstrates consistent performance over more advanced approaches.

We present a comparison between our approach and the baseline method, as well as RCIL [9], under configurations 10-1 and 15-1 on the VOC dataset, as shown in Fig. 4. Initially, our method shows marginal differences compared to the baseline in the early steps. However, after a period of training, our approach progressively outperforms the baseline, indicating its effectiveness in mitigating the forgetting issue.

2) Comparison on ADE20K: The ADE20K dataset presents greater challenges due to its higher number of categories and larger data volume. On the ADE20K dataset, we evaluate our method on PLOP [11], SSUL [13] and CoinSeg [14], as illustrated in Table II. Our approach consistently demonstrated improvements across the 100-50 (2 steps), 100-10 (6 steps), 100-5 (11 steps), and 50-50 (3 steps) configurations in all three methods. This indicates that our method is also suitable for large-scale datasets.

3) Visualization: The experimental results are visually represented in Fig. 5 and Fig. 6, displaying the results of each step

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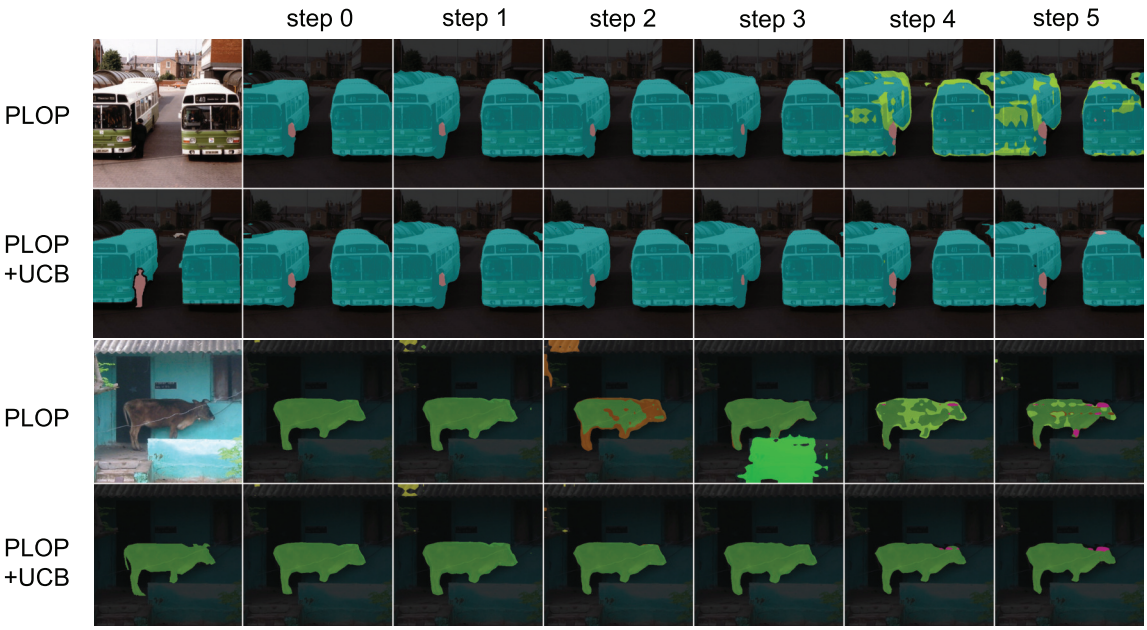


Fig. 5. Visualization comparison on Pascal-VOC 15-1 scenario with PLOP [11]. We showcase the evolving results of the same image throughout the continual learning process. Our approach exhibits remarkable stability throughout the training phase, remaining impervious to the introduction of new classes that could potentially induce semantic drift.

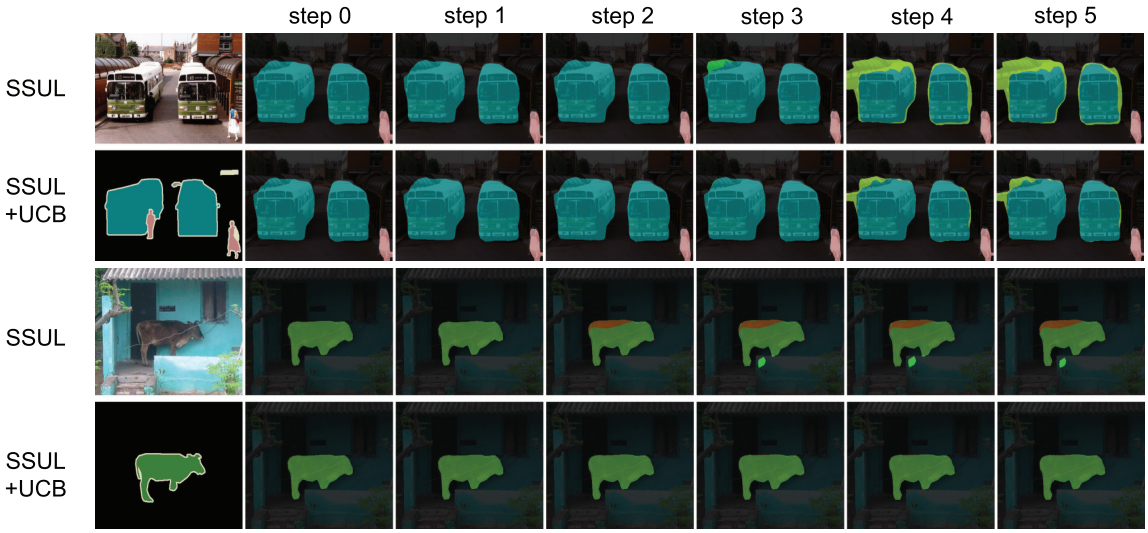


Fig. 6. Visualization comparison on Pascal-VOC 15-1 scenario with SSUL [13]. We present the evolving results of the same image throughout the continual learning process. Our approach demonstrates remarkable stability during the training phase, remaining impervious to the introduction of new classes that could potentially induce semantic drift.

using the PLOP and SSUL with 15-1 configuration. We visualize more complex scenarios containing additional objects, as shown in Fig. 7, where our method demonstrates stable performance. Upon comparison, our method demonstrates superior stability throughout the training process, remaining unaffected by the introduction of new classes that may lead to semantic drift, and showing less susceptibility to erroneous pseudo-labels.

4) Failure Cases Visualization and Analysis: We visualize the failure cases in the Pascal-VOC 15-1 scenario with SSUL [13], as shown in Fig. 8. In the first image, the classes "cow" and "cat" are incorrectly classified as "dog" due to their visual similarity. In the second image, the introduction of the new class "train" leads to some instances of "bus" being misclassified as "train". These examples highlight the challenge of distinguishing between similar classes, which results in category confusion, especially during the incremental introduction of new classes. Therefore, effectively handling the issue of similar classes in the incremental learning process remains a significant challenge and an area for future work.

C. Ablation Study

To validate the effectiveness of our approach, we conduct ablation experiments on the two proposed weights:

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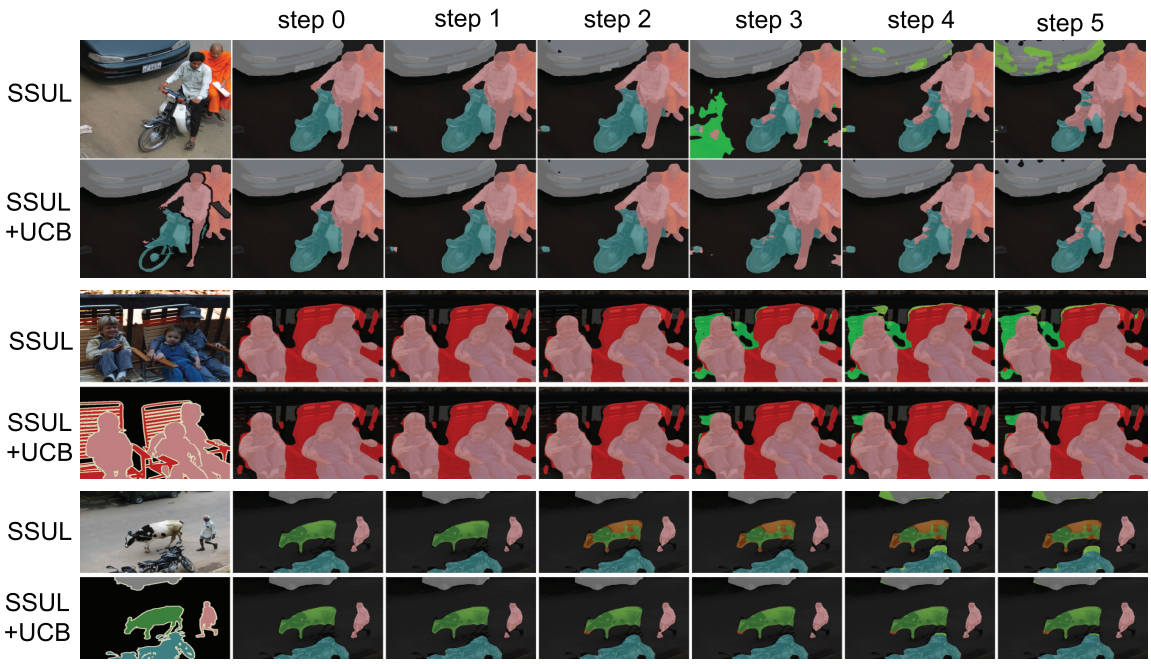


Fig. 7. Visualization comparison on the Pascal-VOC 15-1 scenario with SSUL [13], including more complex scenes with multiple objects.

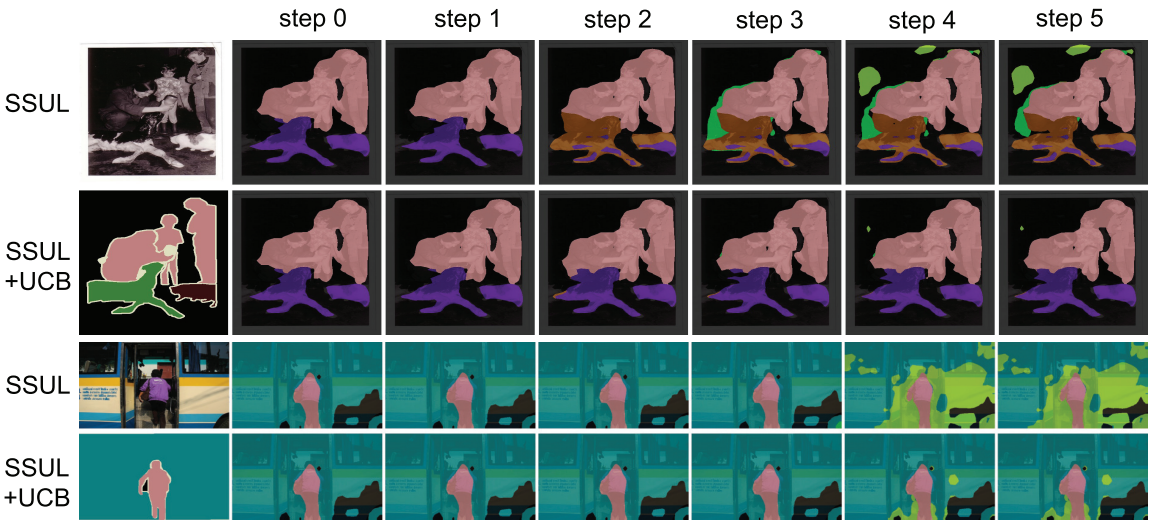


Fig. 8. Failure cases in the Pascal-VOC 15-1 scenario with SSUL [13]. In the first image, the classes "cow" and "cat" are incorrectly classified as "dog" due to their visual similarity. In the second image, the introduction of the new class "train" causes some instances of "bus" to be misclassified as "train".

uncertainty-aware weights and class balance weights. Furthermore, we empirically examine the impact of hyperparameters including filter threshold E_0 and uncertainty scale λ in Eq. (5).

1) Component Ablations: For better observation of the distinctions between our class balance strategy and the conventional approach of reducing new class weights to prevent class confusion, we include the reduction of new class weights strategy [12] as our baseline. The baseline also incorporates the strategy of filtering unconfident pixels. Ablation experiments were conducted on PLOP and CoinSeg under the Pascal-VOC 15-1 overlapped setting. Uncertainty-aware denotes the exclusive use of weights from Section IV-A. Class balance indicates the exclusive use of weights from Section IV-B. UCB is a combination of these two sets of weights.

As shown in Table IV. The uncertainty-aware weights effectively mitigate forgetting by slowing down the impact of erroneous pseudo-labels on the model. Additionally, it can also enhance the learning of new classes. In comparison to the strategy of solely reducing new class weights, class balance weights can dynamically adjust the balance among different classes, resulting in superior performance compared to the baseline. By integrating both approaches, UCB achieves better overall performance than using them individually.

2) Scale Factor Selection: We discussed the hyperparameter λ in Eq. (5). To prevent the variation of E_0 from affecting the overall uncertainty-aware weights, we scaled the weights by scaling the threshold to $[1, \lambda e]$, where different λ represent the weight magnitudes. Although we eventually normalized the weights, a larger λ resulted in higher weight variance

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TABLE III
RESULTS OF MIOU (%) AT THE FINAL STEP ON THE PASCAL-VOC DATASET FOR THE 5-3 SETTING

Method	Old	New	All
MiB [62]	57.10	42.56	46.71
RCIL [9]	63.10	34.60	42.80
BARM [49]	71.30	57.00	61.10
PLOP [11]	25.70	30.00	28.77
+UCB (Ours)	38.82	35.36	36.35
SSUL [13]	72.44	50.67	56.89
+UCB (Ours)	72.54	52.42	58.17
CoinSeg [14]	78.15	70.51	72.69
+UCB (Ours)	78.52	71.64	73.61

TABLE IV
ABLATION STUDY ON PASCAL-VOC 15-1 SETTING. WE INCLUDE THE REDUCTION OF NEW CLASS WEIGHTS STRATEGY [12] AS OUR BASELINE

Method	**Components**	**0-15**	**16-20**	**All**
PLOP	Baseline	69.16	6.85	54.32
	Uncertainty-aware	70.31	15.03	57.15
	Class balance	71.56	8.87	56.64
	+UCB	71.83	15.43	58.40
CoinSeg	Baseline	82.72	52.09	75.43
	Uncertainty-aware	83.59	53.51	76.42
	Class balance	83.19	55.66	76.64
	+UCB	83.28	57.45	77.14

TABLE V
ABLATION STUDY OF λ ON PLOP AND COINSEG IN THE 15-1 AND 10-1 SETTINGS

Method	λ	0.5	0.75	1	1.5	2
PLOP	15-1	56.27	56.61	58.40	57.71	58.34
	10-1	37.38	37.52	36.28	36.81	36.23
	Avg.	46.83	47.07	47.34	47.26	47.28
CoinSeg	15-1	76.12	76.49	77.14	76.44	75.35
	10-1	72.93	72.84	72.84	73.42	72.79
	Avg.	74.53	74.67	74.99	74.93	74.07

and greater uncertainty influence. As shown in Table V, we conduct tests on PLOP and CoinSeg with different λ values [0.5, 0.75, 1, 1.5, 2] on 15-1 and 10-1 settings, obtaining the average mIoU on the two tasks as the evaluation metric. It can be observed that the value of $\lambda = 1$ achieved satisfactory results considering both settings. We chose $\lambda = 1$ as the parameter for all experiments.

3) Filter Factor Selection: We discuss the hyperparameter E_0 in Eq. (5). Acting as the threshold for filters, E_0 is used to set the pixel weights with entropy greater than E_0 to 0, completely filtering out these uncertain pixels. Experimental validations were conducted using PLOP and CoinSeg on settings 15-1 and 10-1, with the average mIoU of the three experiments serving as the evaluation standard, as shown in Table VI. We observe that different models, due to variations in performance, require different optimal values of E_0. Models with better performance tend to require lower values of E_0.

TABLE VI
ABLATION STUDY OF E_0 ON PLOP AND COINSEG IN THE 15-1 AND 10-1 SETTINGS

Methods	E_0	0.3	0.5	0.8	1
PLOP	15-1	56.27	58.27	58.40	56.74
	10-1	29.67	34.17	36.28	37.81
	Avg.	42.97	46.22	47.34	46.99
CoinSeg	15-1	76.12	77.14	76.58	76.42
	10-1	72.80	72.84	72.67	72.75
	Avg.	74.46	74.99	74.63	74.59

Based on the performance of different methods, we select $E_0 = 0.8$ for PLOP and $E_0 = 0.5$ for SSUL and CoinSeg.

D. Computational Cost Analysis

Our proposed method introduces loss reweighting during the training phase, which incurs minimal computational overhead (94.938 GFLOPS for the model's forward pass compared to just 0.024 GFLOPS for our additional compute cost) and does not require any additional learnable parameters. During inference, the computational process remains consistent with the original method. In terms of resource consumption, the only additional requirement is a tensor that scales with the number of classes to track class ratios, resulting in negligible storage overhead. Unlike methods such as ST [58], which rely on iterative Gaussian fitting algorithms executed on the CPU, our approach relies solely on basic arithmetic operations—addition, subtraction, multiplication, and exponentiation—ensuring enhanced computational efficiency.

E. Comparison With Other Uncertainty Methods

We also compare our method with uncertainty approaches from other domains, such as ST [58], which models uncertainty using two Gaussian distributions, and URN [57], which employs scaling operations with CRF to simulate the true behavior of noisy pseudo-masks. Additionally, MBS [48] filters foreground pixels that may be confused with background classes by comparing the probabilities of foreground and background. We conducted experiments on CoinSeg, with the results presented in Table VII. These methods all help mitigate catastrophic forgetting of old categories. Our method shows a consistent performance advantage over the others. Notably, methods like ST and URN require substantial additional computation and time for Gaussian fitting and CRF iterations, whereas our method avoids such complex processing.

F. Effects of Wrong Pseudo-Labels

To investigate the impact of erroneous pseudo-labels on continual learning, we conduct experiments on the more challenging 10-1 task, where the model faces greater forgetting challenges. We examined the model's performance under two scenarios: using Ground Truth labels (GT) which means no erroneous pseudo-labels and deleting wrong pseudo-labels (DW), as shown in Table VIII. We find that removing erroneous pseudo-labels can exacerbate class imbalance, so we introduce class balance weights (CB) proposed in Section IV-B to alleviate this issue. Results indicate a significant

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TABLE VII

PERFORMANCE COMPARISON OF VARIOUS UNCERTAINTY METHODS ON COINSEG [14]. THE TRAINING COST REFERS TO THE TOTAL TRAINING TIME ACROSS ALL INCREMENTAL STAGES ON 15-1, USING TWO 3090 GPUS

Method	19-1 (2 steps)			15-5 (2 steps)			15-1 (6 steps)			10-1 (11 steps)			Training Cost
	Old	New	All	Old	New	All	Old	New	All	Old	New	All	
Baseline	81.14	37.70	79.07	83.29	67.00	79.41	81.73	55.30	75.44	77.98	61.48	70.13	3.5h
+ST [58]	82.29	19.62	79.31	83.09	67.67	79.42	82.45	51.64	75.00	79.65	60.68	70.61	14.5h
+URN [57]	81.49	27.42	78.91	83.19	67.98	79.57	82.68	52.34	75.46	79.72	62.86	71.69	20h
+MBS [48]	82.49	21.28	79.57	83.04	67.88	79.43	82.19	49.90	74.51	78.38	**63.61**	71.34	3.5h
+Ours	**82.82**	**56.73**	**81.58**	**83.37**	**68.91**	**79.92**	**83.47**	**54.73**	**76.63**	**79.90**	63.51	**72.09**	3.5h

TABLE VIII

EFFECTS OF WRONG PSEUDO-LABELS. GT REFERS TO THE USE OF GROUND TRUTH LABELS, WHILE DW INDICATES THE DELETION OF WRONG PSEUDO-LABELS. CB DENOTES THE APPLICATION OF CLASS BALANCE WEIGHTS

Method	Old	New	All
Baseline	44.03	15.51	30.45
GT	50.24	25.66	38.54
GT+CB	57.04	31.01	44.65
DW	20.79	10.47	15.88
DW+CB	55.11	20.82	38.78
UCB (Ours)	50.76	20.36	36.28

TABLE IX

THE INFLUENCE OF THE MAGNITUDE OF CLASS BALANCE WEIGHTS ON PERFORMANCE IN THE PLOP AND COINSEG 15-1 SETTING

Methods	PLOP			CoinSeg		
β	old	new	all	old	new	all
0.25	71.27	18.83	58.79	82.78	55.13	76.20
0.5	71.72	17.96	58.92	83.25	56.07	76.67
1	71.83	15.43	58.40	83.28	57.45	77.14
2	72.37	8.06	57.06	82.90	54.54	76.20
4	72.64	5.14	56.57	82.71	51.46	75.27

TABLE X

THE INFLUENCE OF THE MAGNITUDE OF CLASS BALANCE WEIGHTS ON PERFORMANCE IN THE PLOP AND COINSEG 10-1 SETTING

Methods	PLOP			CoinSeg		
β	old	new	all	old	new	all
0.25	39.23	12.18	26.35	80.07	65.15	72.96
0.5	46.79	19.12	33.61	80.35	66.22	73.63
1	50.76	20.36	36.28	79.72	65.28	72.84
2	54.69	19.88	38.11	78.92	64.07	71.84
4	54.82	16.37	36.52	77.91	63.45	71.03

performance improvement when using Ground Truth labels, resulting in reduced forgetting and facilitating new class learning. Moreover, deleting wrong pixels also shows substantial performance gains when utilizing class balance weights. By reducing the weight of high-uncertainty pixels, our proposed method effectively mitigates the negative impact of erroneous pseudo-labels on the model.

G. Impact of the Magnitude of Class Balance Weights

We investigate the influence of the min and max values in Eq. (7) on model performance. Consistent with common practice [12], we bounded the range of the class balance weight, denoted as α, between $\sqrt{\frac{C^t}{C^{1:t-1}}}$ and $\sqrt{\frac{C^{1:t-1}}{C^t}}$. The range of the class balance weight determines the level of importance assigned to class balance. To facilitate discussion, we introduce a coefficient β as a metric for the range of the class balance weight α, where a higher β signifies a more pronounced difference in weights related to class balance. With the introduction of β, the min and max values can be represented as $\sqrt{\frac{1}{\beta}\cdot\frac{C^t}{C^{1:t-1}}}$ and $\sqrt{\beta\cdot\frac{C^{1:t-1}}{C^t}}$, respectively. However, a higher β does not always lead to superior results, as excessively low weights may hinder the model's capacity to learn from new classes. We choose experimental values for β as [0.25, 0.5, 1, 2, 4]. The experiments are executed using the PLOP and CoinSeg with the configurations of Pascal VOC 15-1 and 10-1, as detailed in Table IX and Table X. As β increases, it limits the model's ability to assimilate new classes due to reduced weights assigned to new classes. Various values of β have distinct impacts on the sequences of individual tasks and the performance of different methods. In our experiment, we utilize a default value of $\beta = 1$ on the Pascal VOC dataset, which is not the best but still performs well across all configurations.

VI. LIMITATION AND CONCLUSION

In this paper, we have pinpointed two frequently neglected yet pivotal concerns within CSS: the assessment of pseudo-labels and the challenge of class balance. We propose UCB that reallocates loss weights based on uncertainty, allowing for flexible adjustments of class weights by statistically analyzing pixel-level class information. Our approach, while simple, has demonstrated excellent performance across various tasks.

Nevertheless, it is essential to acknowledge several limitations of our approach. Our statistical analysis of class information is based on pseudo-labels, which can lead to incorrect class balance weights and assign higher weights to mislabeled instances. While we address this issue by incorporating uncertainty-aware weights, it still has a notable impact on the model's performance. Additionally, our strategy for dealing with mislabeled data emphasizes mitigating their influence rather than strictly correcting the errors. As a result, misclassifications cannot be effectively rectified.

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For future research, a key direction is to address the severe issue of class imbalance. This includes tackling class confusion caused by insufficient samples, which may lead to misclassification between new and similar existing classes. From the perspective of prediction uncertainty, future studies could incorporate additional uncertainty reference information or focus on enhancing the model's ability to better differentiate between predictions. We believe that addressing these challenges will open up promising avenues for further research in CSS problems.

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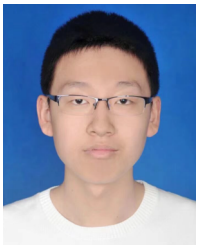
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